

Understanding User Requirements in LLM-Augmented BIM Systems: A TAM-Based Evaluation of NADIA-S

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ABSTRACT

Recent research integrating large language models (LLMs) with building information modeling (BIM) has heightened expectations for overcoming technical barriers limiting BIM adoption. However, user requirements for LLM-augmented BIM systems remain underexplored, challenging their practical implementation. This study investigates adoption factors through expert evaluations of NADIA-S, a speech-based LLM-augmented BIM system using a generalized six-step framework. Fourteen experienced architects performed interactive design detailing and compliance-checking tasks, providing feedback via Technology Acceptance Model (TAM)-based questionnaires assessing perceived ease of use (PEU), perceived usefulness (PU), and behavioral intention (BI). Results indicated high ratings—PEU: 3.75–4.58, PU: 4.08–4.25, BI: 4.25–4.50—with significant correlations among constructs. Qualitative responses emphasized conversational interface preferences, multimodal input capability demands, enhanced AI comprehension of BIM, improved system stability, and greater control over AI-generated outputs. Addressing these areas can significantly improve system reliability, user confidence, and accelerate the practical adoption of LLM-augmented BIM systems.

INTRODUCTION

Large language models (LLMs) have emerged as powerful tools for reshaping interactions between professionals and digital systems in the architecture, engineering, construction, and operation (AECO) industry (Jang et al. 2025). By offering human-like natural language understanding and generation, LLMs present a future in which the technical barriers to adopting complex platforms—such as building information modeling (BIM)—could be substantially reduced. Indeed, numerous conceptual and experimental studies have demonstrated LLMs’

potential to automate various BIM tasks through more intuitive interactions (Chen et al. 2024; Du et al. 2024a, 2024b; Fernandes et al. 2024; Jang et al. 2024; Jang and Lee 2023; Kang and Park 2024; Yin et al. 2023; Yu et al. 2024; Zheng and Fischer 2023).

Despite these advances, critical questions remain about how AECO practitioners actually perceive LLM-augmented BIM systems. While conversational user interfaces may provide immediate benefits—such as effortless model manipulation and instant project data retrieval—ongoing challenges include system reliability, scalability, and alignment with professional standards and workflows. These factors ultimately determine whether LLM-augmented BIM solutions gain widespread acceptance. In particular, there is a need to understand user satisfaction, perception, and the barriers that could deter regular practice.

To address these gaps, this study presents an expert-focused evaluation of “Natural-Language-Based Architectural Detailing through Interaction with AI via Speech (NADIA-S),” a system developed under a generalized LLM-augmented BIM framework (Jang et al. 2024; Lee et al. 2024). The aim is to assess how well such systems meet user requirements and expectations by examining experts’ views on perceived ease of use (PEU), perceived usefulness (PU), and behavioral intention (BI) through the Technology Acceptance Model (TAM) (Davis 1989). By analyzing 14 experts’ feedback from a case application following demonstrations, we investigated how AECO professionals perceive and anticipate using LLM-augmented BIM systems in practice.

The remainder of this paper is organized as follows: The second section discusses the motivation for this study. The third section describes how user requirements and perceptions were evaluated using TAM. The fourth section outlines experts’ evaluations of PU, PEU, and BI, highlighting current challenges and potential improvements. Finally, the fifth section provides final remarks on the findings and proposes directions for future research.

BACKGROUND

Although BIM has long been recognized as a foundational technology for gathering and integrating building project information in AECO projects, its user interfaces typically rely on complex menus and specialized commands. Such complexity often poses adoption barriers (Faisal Shehzad et al. 2022; Won et al. 2013), especially for professionals who lack extensive software training. In contrast, modern LLMs—such as generative pretrained transformer (GPT) (Radford 2018) and large language model Meta AI (LLaMA) (Touvron et al. 2023)—demonstrate near-human capabilities in natural language understanding and generation. These LLMs suggest a new paradigm whereby intuitive text or voice commands could replace cumbersome manual inputs in BIM systems.

Table 1 outlines previous studies integrating LLMs into BIM systems to enable natural language-based task execution. Their objectives include information retrieval (Kang and Park 2024; Zheng and Fischer 2023), compliance checking (Chen et al. 2024; Yu et al. 2024), command execution (Fernandes et al. 2024), primitive BIM model generation (Du et al. 2024a; b), and design detailing (Jang et al. 2024; Jang and Lee 2023).

Table 1. Previous studies integrating LLM into BIM systems.

Objectives	References	Description
Information retrieval	Yin et al. (2023)	Combining Owl2Vec and a graph neural network to parse natural language queries (NLQs).
	Zheng and Fischer (2023)	Utilizing instruction prompting for slot filling for BIM queries.
	Kang and Park (2024)	Applying retrieval augmented generation (RAG) on the industry foundation class (IFC) graph.
Compliance checking	Chen et al. (2024)	Mapping BIM model and ontology, classifying regulation text, and extracting structured information for compliance checking.
	Yu et al. (2024)	Interpreting and modifying compliance checking code through natural language conversation for barrier-free design.
Command execution	Fernandes et al. (2024)	Executing predefined BIM command methods (e.g., change transparency, isolate, hide, set colors, tag, view settings).
Primitive BIM model generation	Du et al. (2024a; b)	Utilizing multiple LLM agents (i.e., product owner, architect, programmer, reviewer) to generate primitive BIM models.
Design detailing	Jang et al. (2024), Jang and Lee (2023)	Generating, modifying, and applying design details to BIM models upon natural language task prompts.

Information retrieval: Yin et al. (2023) proposed a method using Owl2Vec and graph neural networks to parse natural language queries (NLQs), thereby enabling semantic searches within BIM data. Zheng and Fischer (2023) integrated an LLM into BIM systems to enable NLQs through chat messages, employing instruction prompting for slot filling. Kang and Park (2024) applied a retrieval-augmented generation (RAG) approach to IFC data to retrieve BIM object properties required by NLQs.

Compliance checking: Chen et al. (2024) developed a framework that combines LLMs, deep learning-based regulatory text classification, and ontology knowledge models to automatically extract structured data and verify BIM compliance with regulatory requirements. Yu et al. (2024) proposed an interactive system wherein compliance-checking code can be interpreted and updated via natural language conversations, aiding barrier-free design checks.

Command execution: Fernandes et al. (2024) demonstrated how predefined BIM commands—e.g., changing transparency or hiding objects—can be executed through natural language requests, enhancing workflow efficiency.

Primitive BIM model generation: Du et al. (2024b) introduced an LLM agent-based method with predefined toolsets to support primitive BIM modeling. Subsequently, Du et al. (2024a) extended the approach to a multi-agent framework for automated BIM authoring.

Design detailing: Jang and Lee (2023) proposed a method for an LLM to make design decisions by interpreting building context and applying appropriate design details within BIM through LLM chaining. Jang et al. (2024) expanded this capability to include the generation, modification, and application of design details to BIM models via natural language conversations.

Despite these extensive efforts, limited attention has been given to how AECO professionals actually perceive and engage with such systems. Existing studies often focus on technological advancements but overlook key user-centric factors, such as system usefulness, ease of use, and willingness to adopt—all of which are pivotal for driving practical uptake (Davis 1989).

Given that user requirements and perceptions directly influence the success and acceptance of LLM-augmented BIM systems, addressing this gap is essential for ensuring widespread adoption and meaningful integration into professional workflows. Recognizing these issues, this study employs the TAM to examine key factors—PU, PEU, and BI. Experts’ responses to questionnaires and their qualitative feedback on NADIA-S—a speech-based, LLM-augmented BIM system designed for architectural detailing and compliance checks (Lee et al. 2024)—were collected to shed light on how such technologies might be received and further developed.

RESEARCH METHOD

We conducted a TAM-based analysis of PEU, PU, and BI using experts’ responses after they had used NADIA-S, an LLM-augmented BIM system. TAM is a widely recognized framework for understanding and predicting user acceptance of new technologies. According to Davis (1989) two primary factors—PEU and PU—directly shape an individual’s BI to use a system, which then influences actual usage (Figure 1).

Participants: Fourteen experienced architects were directly recruited for the study, with all participants providing informed consent and assurances of confidentiality. Their professional backgrounds were assessed by collecting data on both design and BIM-based design experience, revealing that architects with less than 5 years of experience had 5 design and 6 BIM experiences; those with 5 to 15 years had 4 design and 7 BIM experiences; and those with 15 or more years had 7 design and 1 BIM experience.

Data collection: Data was collected through a structured questionnaire based on the TAM constructs (PEU, PU, BI) as outlined in Table 2. The questionnaire contained 16 items rated on a 5-point Likert scale of “Strongly Disagree” (1) to “Strongly Agree” (5). These items were adapted from established TAM instruments to suit the NADIA-S context. Along with quantitative data, participants contributed qualitative feedback on each TAM construct, offering additional insights into their experiences and expectations.

Procedure: Participants engaged in interactive sessions with NADIA-S to perform architectural detailing tasks. Before the session, they were introduced to the system’s capabilities and shown a demonstration. The features included detail generation and modification of wall objects, applying details to walls, slabs, columns, beams, windows, and doors, and checking design compliance on walls, slabs, columns, and beams (Table 3). After the introduction and

demonstration, each participant had ten minutes to explore NADIA-S's features, emphasizing natural language interactions for BIM detailing. They then completed the TAM questionnaire.

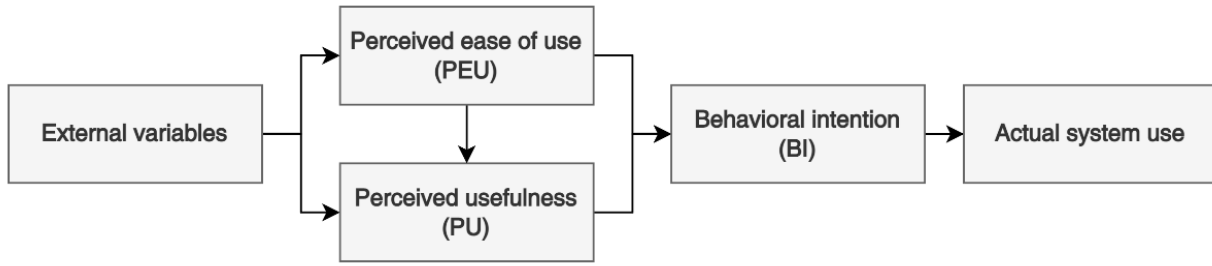


Figure 1. The TAM framework consists of PEU, PU, and BI (David, 1989).

Table 2. TAM questionnaire related to perceived usefulness (PU), perceived ease of use (PEU), and behavioral intention (BI).

TAM Construct	Code	Items
Perceived usefulness (PU)	PU1	Using this system in my job would enable me to accomplish tasks more quickly.
	PU2	Using this system would improve my job performance.
	PU3	Using this system in my job would increase my productivity.
	PU4	Using this system would enhance my effectiveness on the job.
	PU5	Using this system would make it easier to do my job.
	PU6	I would find this system useful in my job.
Perceived Ease of Use (PEU)	PEU1	Learning to operate this system would be easy for me.
	PEU2	I would find it easy to get this system to do what I want it to do.
	PEU3	My interaction with this system would be clear and understandable.
	PEU4	I would find this system clear and understandable.
	PEU5	It would be easy for me to become skillful at using this system.
	PEU6	I would find this system easy to use.
Behavioral Intention (BI)	BI1	I would consider using this system.
	BI2	I plan to use this system.
	BI3	I will continue using this system.
	BI4	I will recommend this system's excellence to others.

Table 3. Overview of NADIA-S Functionalities Tested on Various Object Types.

	Detail generation and modification	Apply details on objects	Reference-based design compliance check
Walls	✓	✓	✓
Slabs		✓	✓
Columns		✓	✓
Beams		✓	✓
Windows		✓	
Doors		✓	

TAM Analysis: The data analysis relied primarily on quantitative measures including mean Likert scale and standard deviation (SD), to summarize responses from fourteen architects (N=14). The mean of 5-point Likert scales for each questionnaire was interpreted by dividing the range (5–1=4) into equal 0.8 intervals ($4 \div 5$). Starting from 1, intervals (e.g., 1.0–1.8, 1.8–2.6, 2.6–3.4, 3.4–4.2, and 4.2–5.0) correspond to levels from “Strongly Disagree” to “Strongly Agree.” The SD was analyzed to assess the variability of responses: a low SD (e.g., <1) indicates a high level of agreement, with responses tightly clustered around the mean; a moderate SD (e.g., 1–1.5) reflects some variability but still exhibit a clear trend; and a high SD (e.g., >1.5) indicates diverse opinions, with responses spread widely across the scale. Additionally, a qualitative analysis of the participants’ comments on each factor was conducted by classifying the responses into “positive,” “neutral,” and “negative” categories and identifying specific perceived advantages and future requirements of LLM-augmented BIM systems.

RESULTS

An analysis of user responses (N=14) on a 5-point Likert scale (Figure 2) demonstrates generally positive perceptions of LLM-augmented BIM systems.

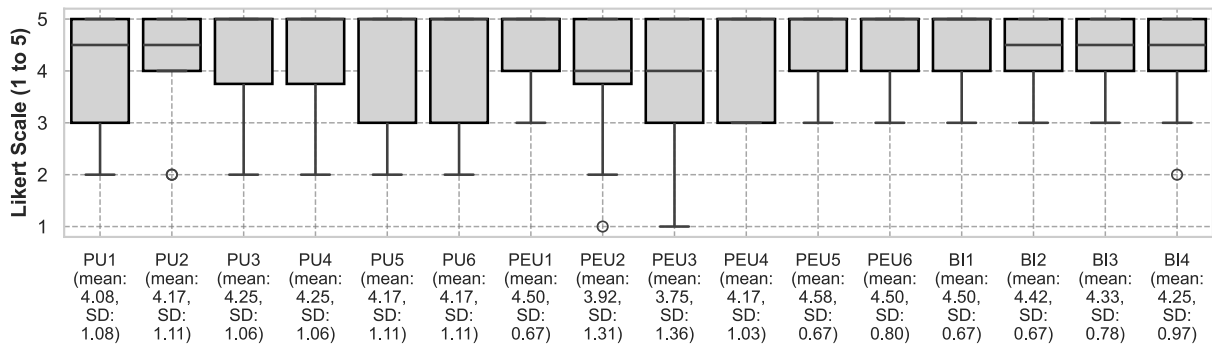


Figure 2. Distribution of responses on questionnaires.

For PU, mean scores ranged from 4.08 to 4.25 (i.e., between “Agree” and “Strongly Agree”). Two items—“PU3” (increased productivity) and “PU4” (enhanced job effectiveness)—both scored 4.25, corresponding to “Strongly Agree,” thereby suggesting that participants foresee meaningful benefits for productivity and effectiveness. The SDs for PU (1.06–1.11) indicated moderate variability yet still showed a consistently positive trend. For PEU, mean scores fell between 3.75 and 4.58. Notably, PEU1, PEU5, and PEU6 scored 4.50, 4.50, and 4.58, respectively, suggesting a strong consensus that the system is easy to learn and operate. In contrast, PEU2 and PEU3 exhibited lower mean values (3.92 and 3.75) and higher SDs (1.31 and 1.36), indicating that some participants experienced occasional difficulties in performing desired tasks or fully understanding system interactions. Regarding BI, the mean scores for BI1, BI2, BI3, and BI4 were 4.50, 4.42, 4.33, and 4.25, with all SDs under 1. These results suggest that participants are inclined to continue

using LLM-augmented BIM systems in professional settings. Overall, the quantitative analysis reveals a favorable perception of the system’s usefulness, ease of use, and adoption potential.

To gain further insight into participants’ satisfaction and requirements, a qualitative analysis of their comments was conducted for each TAM factor (Table 4). The findings are summarized below.

Table 4. Types of comments.

TAM Factor	Sentiment	Types	Number of comments
Perceived Ease of Use (PEU)	Positive	Intuitiveness and convenience of verbal interaction	6 (42.86%)
		Advantages of conversational user interface	3 (21.43%)
		Convenience of bulk modifications	2 (14.29%)
	Neutral	Need for improvement in system stability	1 (7.14%)
		Preference for text input in certain environments (i.e., office environments)	1 (7.14%)
		Request for Excel import features	1 (7.14%)
Perceived Usefulness (PU)	Positive	Time savings through design automation	3 (21.43%)
		Exhibit potential for extended capabilities through LLM augmented Revit add-in.	3 (21.43%)
		Ability to execute design tasks required in practice	3 (21.43%)
	Neutral	Expectations for advanced Revit features	2 (14.29%)
		Needs to comprehend existing functions (e.g., basic commands, primitive BIM object generations)	3 (21.43%)
Behavioral Intention (BI)	Positive	Practical applicability in real-world tasks	2 (14.29%)
		Time-saving advantages	2 (14.29%)
		Overall convenience and functional benefits	2 (14.29%)
	Neutral	Need for further expansion and specialization (i.e., legal review, expert systems, customization for a project)	4 (28.57%)
		System stability and ongoing updates	3 (21.43%)
	Negative	Insufficient stabilization	1 (7.14%)

Preferences on interactive conversational user interface: A large share of the PEU-related comments (11, or 78.57%) were positive, highlighting that verbal commands feel faster and more convenient than manual inputs. Specifically, six responses (42.86%) emphasized the intuitiveness of voice-based interaction, while three (21.43%) mentioned the advantages of a dialogue-driven approach. Although these comments point to a system appropriate for professionals less familiar with complex BIM processes, some users (14.29%) still experienced occasional crashes. Moreover, concerns arose about accountability for design outputs generated by AI. Several participants thus recommended clear feedback and revision pathways, allowing users to validate or modify automatically produced changes.

Requirements for multimodal input capability: Many participants advocated for the system to integrate additional data formats beyond natural language—particularly images, tables, and textual specifications. For instance, a PEU-neutral remark explicitly requested Excel import features, underscoring how spreadsheets remain a common data format for project requirements,

scheduling, and cost management. Offering robust multimodal data handling would broaden the system's applicability across varied AECO practices.

Requirements for improving AIs' comprehension of BIM systems: While NADIA-S primarily targets design detailing and compliance checks (Table 3), several participants expressed interest in expanding its scope to encompass a broader range of BIM commands and tasks. Under PU, 21.43% of comments recognized the potential for extending the system's capabilities beyond existing functions, and an equal percentage highlighted the importance of understanding fundamental BIM processes (e.g., basic commands and object generations). These remarks underscore the need for AI to develop a deeper domain-specific knowledge of BIM systems, allowing it to reliably parse user intents and execute both routine and advanced operations. By enhancing its comprehension of object-oriented programming hierarchies and entities regarding pre-defined classes and methods of Revit, the AI could evolve into a more versatile BIM assistant.

Requirements for improving system stability and control over AI-generated results: During testing, some participants experienced minor system instability (e.g., unexpected terminations or unintended command executions). Although these issues may partly reflect the current prototype's limitations, they illuminate a broader challenge in LLM-augmented BIM solutions: determining how much freedom the AI should have versus how much control the user should retain. Responses indicated that domain-specific knowledge could help the AI better interpret user intentions—especially when tasks are more specialized—and that providing transparent oversight of generated outcomes is crucial for professional acceptance. In short, balancing flexibility, reliability, and user control emerged as a key research and development priority.

In summary, participants viewed LLM-augmented BIM as a promising avenue for reducing repetitive work, offering an intuitive interface, and potentially boosting productivity. However, they consistently underscored the importance of multimodal data handling, refining basic BIM commands, and fortifying system stability. Such improvements would help ensure users can verify and control AI-generated results with confidence, ultimately enhancing the broader acceptance and efficacy of LLM-driven BIM solutions in the AECO industry.

CONCLUSION

This study assessed how an LLM-augmented BIM system, exemplified by NADIA-S, is perceived by fourteen experienced architects (N=14) through the lens of the TAM. Survey findings on a 5-point Likert scale showed that participants rated the system highly for both PEU and perceived usefulness PU, with PEU scores ranging from 3.75 to 4.58 and PU scores from 4.08 to 4.25. BI also scored favorably, at 4.25 to 4.50, indicating participants' intention to adopt the LLM-augmented BIM approach in their professional workflows.

Qualitative feedback similarly highlighted the promise of LLM-augmented BIM technology, with participants noting how a conversational interface could simplify complex tasks and reduce repetitive work, especially for professionals less familiar with BIM systems. However,

the comments also pointed to areas requiring further improvement. Several users requested multimodal data input so that practical documents—including spreadsheets and images—could be directly integrated into automated BIM workflows. Others underscored the need to deepen the AI’s understanding of fundamental BIM processes beyond the system’s current focus on design detailing and compliance checks. They further emphasized the importance of enhancing stability and transparent oversight, citing occasional crashes, unintended commands, and the need for clearer accountability and verification of AI-generated outcomes.

While previous studies have focused on the technological development of LLM-augmented BIM, this study contributes to the body of knowledge by identifying user requirements for LLM-augmented BIM. Through expert evaluations, it sheds light on these critical yet underexplored dimensions, offering insights into the user requirements and proposing clear directions for their future development and practical application. While the findings are encouraging, the reliance on a relatively small sample of fourteen participants limits the generalizability of the results. Future research involving larger, more diverse samples would help confirm these observations and clarify how varying roles or project contexts might influence PEU, PU, and BI. Additional statistical analyses—such as advanced regression models or mediation tests—could also refine understanding of the TAM constructs concerning user perceptions. By addressing these priorities and leveraging the latest advancements in LLM technology, these systems can further enhance productivity, streamline workflows, and ultimately reshape the adoption of BIM in the AECO industry.

ACKNOWLEDGEMENTS

This work is supported in 2025 by the Korea Agency for Infrastructure Technology Advancement (KAIA) grant funded by the Ministry of Land, Infrastructure and Transport (Grant RS-2021-KA163269 and RS-2024-00407028) and the Hans Fischer Senior Fellowship program at the Technical University of Munich - Institute for Advanced Studies (TUM-IAS) in Germany.

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